# SIMATS ENGINEERING
## ASSESSMENT TOOL 1

**COURSE NAME: CSA07 COURSE CODE:Computer Networks**

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**COURSE OUTCOME COVERED**

***CO1:*** *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

*Assessment Tool Weightage: 40%*

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Annexure A

**SHORT ANSWER TEST**

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***Code of Conduct:***

*I* K. Sakthivel [192521484] *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:*** K Sakthi

| Criteria | Weightage | Q1 10marks | Q2 10marks | Q3 10marks | Q4 10marks |
|---|---|---|---|---|---|
| Understanding of Concepts | 25% | 2 | 2 | 2 | 2 |
| Explanation | 25% | 1 | 1 | 1 | 1 |
| Application | 20% | 1 | 1 | 1 | 1 |
| Organization | 15% | 1 | 1 | 1 | 1 |
| Accuracy | 10% | 1 | 1 | 1 | 1 |
| Clarity | 5% | 1 | 1 | 1 | 1 |

**Total Marks** 28 **/40**

***Signature of the Faculty***

## Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests

of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

**9.**A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

**10.**A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

**RUBRICS FOR CALCULATION**

| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Understanding of Concepts | 25% | Demonstrates a thorough, accurate understanding of concepts | Good understanding with minor gaps | Basic understanding with some misconceptions | Poor understanding; major misconceptions |
| Explanation | 25% | Detailed, well-explained answers with relevant examples | Clear explanation with adequate details | Limited explanation; lacks depth | Poor or unclear explanation |

| | | | | | |
|---|---|---|---|---|---|
| Application | 20% | Effectively applies concepts with strong analytical ability | Applies concepts with minor errors | Limited application with weak analysis | No application or incorrect analysis |
| Organizatio n | 15% | Well-structured answers with logical flow and coherence | Organized with minor issues in flow | Basic structure, lacks clarity | Poor organization, incoherent answers |
| Accuracy | 10% | Completely accurate and covers all required points | Mostly accurate with minor omissions | Partially correct with missing points | Incorrect answers with major gaps |
| Clarity | 5% | Clear, neat, professional presentatio n | Understanda ble with minor issues | Limited clarity, readability issues | Poorly written, difficult to understand |

CO1 - AT - 1

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CSA0709 Computer Networks 192521484

20/10/2026

Short Answer test

Dr. R. Shanker

1. A Company needs to transfer a 150 MB design file from a client computer to a remote server over a network with a bandwidth of 50 mbps. The Transport layer divides the file into 1500 - byte segements while the Data link layer encapsulates each segment into frams by adding 40 bytes of header and trailer. Calculate the total number of segment and frames transmitted

2. A Transport Layer protocal uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, Calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted.

3. A 12,000 - byte IP packet is transmitted through a router where the maximum transmission unit (MTU) is 1500 bytes Assuming each fragment requires a 20 byte IP header. Calculate the total number of fragement created and the total header overhead introduced.

4. A Transport layer protocol uses a sliding window size of 8 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames. Calculate the number of transmission window required. If one frame in every window is lost and must be retransmitted. determine the total number of retrans Explain how flow control improves Communication efficiency.

1.

Ans:

Given

Begin file = 100 MB

Bandwith = 150 MBPS

Transfer layer = 1500 bytes.

* 150 MB = $150 \times 10^6$ bytes

Segment = $(150 * 10^6) / 1500 = 100,000$ Segment

* Number of frame

one frame per segment : 100,000 frames

* Data link overhead

overhead = $100,000 \times 40 = 4000,000$ bytes = 4 MB

Transmission time

Total transmitted = $150,000,000 + 4,000,000 = 154,000,000$ byt.

Total bits = $154,000,000 \times 8 = 1232,000,000$ bits

Time = 1,232,000,000 / 50,000,000 = 24.64 second.

Layer function :

Transport layer : Segment data and may use sequence number, acknowledgement, and retransmission depending on the protocol

Data link Layer : frames data, provides link addressing, error detection and may retransmit damaged frames on supported links.

Given

window size : 6 frames

Frames carries : 1024 bytes

Sender transmits = 48 frames

Formula:

Number of transmission = $\frac{\text{Total frames}}{\text{window size}}$

$= \frac{48}{6} = 8$

Retransmission = Number of window

$= 8 \times 1 = 8$

Total user data = 48 × 1024

= 49,152 bytes.

Flow Control:

Allows multiple frame to be sen
before waiting for acknowlegment

Helps prevent reciver buffer overflo

Improve a transport layer protocol
a sliding window

Supports recovery from lost frames
through acknowlegement and retransmission when th
protocol supports them.

Conclusion: Sliding window flow control improves
communication efficiency by allowing multiple frame
to be sent continuously

3.

Given

IP packet : 12,000 - byt

MTU = 1500 byte

IP per progement : 20 bytes.

Fragment : data per full size.

1500 - 20 = 1480 bytes

Number of fragments :

Assuming 12000 bytes includes the original 20 bytes IP header, original payload :

IP header payload : 12000 - 20

= 11,980 bytes

8 Full fragement Carry

8 x 1480 = 11840 bytes

Remaining payload = 11980 - 11840 .

= 140 bytes

∴ Total fragement = 9 fragement

Header overhead

9 x 20 = 180 bytes

Network Layer function :

Router packet using logical IP address is transmitted through a router .

* MTU is 1500 bytes assuming each fragement requires a 20 bytes IP header supports recovery from last frames through acknowlegement and retransmission

Given:

File Size = 600 MB,

interval = 60 MB,

failure = 390 MB

Checkpoint before failure :

390/60 = 6 checkpoint at 60, 120, 180, 240, 300 and 360 MB

Data to retransmit:

Last Completed checkpoint = 360 MB

390 - 360 = 30 MB

Importance of Synchronization :

Manages Communication Session between application.

Creates synchronization point during long transfer.

Helps resume from saved progress after failure.

Allows multiple frame to be sent before waiting for acknolegment.

A company needs to transfer a design file from a client computer to a remote server over a network.